PyClimaExplorer

Turning Climate Data into Actionable Insights

~ Team Avengers

The Challenge

Climate Data is Complex

Researchers and the public struggle to interpret raw climate datasets from NetCDF files. Temperature, precipitation, and wind speed data remain locked in technical formats.

Current Tools Fall Short

Existing platforms require technical expertise and lack interactive features. Without intuitive visualization, critical climate trends remain invisible to decision-makers.

Our Solution

Interactive Global Heatmap

Visualize temperature, precipitation, and wind patterns across the entire planet with dynamic color coding

Time-Series Trends

Track climate variables at specific locations over decades with interactive line graphs

AI-Powered Insights

Generate automated analysis and detect patterns in climate data instantly

Powerful Features

Global Climate Heatmap

Dynamic visualization of temperature, precipitation, and wind speed across all continents with real-time data rendering

Location-Based Time-Series

Analyze climate trends at any coordinate point with historical data spanning decades

AI-Generated Insights

Machine learning algorithms identify anomalies, trends, and correlations automatically

Climate Comparison Mode

Side-by-side analysis of different time periods (e.g., 1990 vs 2020) to visualize climate change progression

Future Predictions

Project climate variables forward using trained machine learning models

Technology Stack

Frontend

HTML, CSS, Vanilla JavaScript

Creates the interactive user interface for the climate dashboard.

Backend

Python with Flask/FastAPI

Handles API requests and processes climate data.

Data Processing

Pandas and NumPy

Used for cleaning, transforming, and analyzing climate datasets.

Machine Learning

Scikit-learn

Implements prediction models for climate trend analysis.

Visualization

Plotly.js /Chart.js

Generates interactive charts like temperature trends and precipitation graphs.

Dataset

Global Climate Dataset

Includes temperature, precipitation, wind speed, humidity, latitude, longitude, and time date.

Impact & Future Vision

Accelerate Research

Researchers explore climate datasets in minutes instead of hours

Public Understanding

Makes climate trends accessible to non-technical audiences

Data-Driven Decisions

Empowers policymakers with actionable climate insights

Future Enhancements

- 3D globe visualization with WebGL rendering
- Real-time satellite data integration
- AI-driven climate risk prediction for cities
- Mobile application for field researchers

Thank You

We appreciate your time and interest PyClimaExplorer. Together, let's turn data into a brighter, more sustainable future.